AN ECLECTIC TECHNOLOGIES FOUNDATIONAL DEFINITION

FD-LD — Layer Decomposition and Determinism

Foundational Definition: The Two-Layer Architecture, the Determinism Boundary as a Scope Gate, and the Machine-Learning Carve-Out

Version 0.5 — DRAFT · 15 October 2026

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SUGGESTED CITATION

T. Roxey, "FD-LD — Layer Decomposition and Determinism," Version 0.5, in *FD — Foundational Definitions, November 2026 Edition*, Eclectic Technologies, October 2026. https://doi.org/10.5281/zenodo.23401272

This version is a member of FD — Foundational Definitions, November 2026 Edition (https://doi.org/10.5281/zenodo.23401272). The edition is a dated coherence claim: it states the family's state as of 19 October 2026 and is not falsified by a later revision. The earlier October 2026 Edition (https://doi.org/10.5281/zenodo.23117612), September 2026 Edition (https://doi.org/10.5281/zenodo.23000686) and August 2026 Edition (https://doi.org/10.5281/zenodo.22181029) remain citable for what they contain. Version state is authoritative in FD-REG, whose dated copy accompanies this edition. All editions: https://doi.org/10.5281/zenodo.22181028.

Designation: FD-LD
Family: Foundational Definitions
Version: 0.5 — DRAFT (issued in the November 2026 Edition; supersedes v0.4)
Date: 15 October 2026
Supersedes: *Layer Decomposition and Determinism Terminology Annex* v0.1
Author: Timothy E. Roxey
Organization: Eclectic Technologies
Audience: Asset owner/operators; ICS vendors and suppliers; regulatory staff and standards-community reviewers
Parent: None. This instrument defines. See §2.
Companion root definitions: FD-BL, which owns the placement criterion; and FD-EV, FD-BR and FD-LS. FD-BL §4.4 cites this instrument's §4.3 for the scope rule. FD-LD supplies the determinism and layer vocabulary and states the scope rule; it does **not** own placement.

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1. Purpose and the Defect This Instrument Corrects

The annex this instrument replaces did one thing right and one thing wrong, and the two are separable.

Right: it unified two vocabularies. The Eclectic Technologies governance work describes one determinism boundary in two languages — the ISA / design-basis vocabulary of infrastructure and cognitive layers, and the determinism vocabulary of deterministic and non-deterministic systems. Stating that these are the same boundary, so a reader arriving from either recognizes the other, is correct and useful. That unification is retained here in full (§3).

Which documents speak which vocabulary. One family — the ISA Parent Framework, the IEEE V&V Cognitive AI Framework, the sector application guides, and the quality assurance instruments derived from them — speaks of an infrastructure layer and a cognitive layer. Another family — the Command Broker cross-sector framework and its sector annexes, the Command Broker Implementation Guide, *Generative AI Risk in Industrial Control Systems*, and the Design Basis Accident instruments — speaks of deterministic and non-deterministic systems. **Both vocabularies remain in use and neither family is required to adopt the other's terms**; this instrument supplies the equivalence and cedes placement to FD-BL.

Wrong: it fused that boundary with the placement criterion. The annex's mapping table carried a third row equating "the boundary between the layers" with "the Bright Line placement criterion," resolving both to "whether a system's behaviour can be bounded by proof," and asserting that "everything downstream — Command Broker interposition, assessment method, tier rigour — follows from which side a component falls on." That row makes determinism the criterion for *where the line falls*. FD-BL §4.2 and §4.4 hold the opposite: determinism decides **scope** — which components are subject to Bright Line analysis — and **consequence** decides **placement** — where the line falls for the actions a component produces.

The consequence of the fusion was not local. Because the annex was the cited authority for the sector series (issued under the ISA Parent Framework consistency protocol), the fused row propagated as doctrine: DBA-DC-L1 §3.5, DBA-EN-SF, and DBA-ES were found to state placement on the determinism boundary, and the Command Broker Framework §3.2 carried both the fused doctrine and, contradicting it, the correct consequence criterion. This instrument withdraws the fused row and thereby removes the doctrine at its source.

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2. Authority, Family, and Convention

2.1 Precedence

FD-LD owns, for the whole corpus, the two-layer decomposition, the determinism vocabulary, and the machine-learning carve-out — that is, the **scope gate** that decides which components are subject to Bright Line analysis and the Command Broker obligation. The governing statement of that scope is at §4.3.

FD-LD does **not** own placement. Where the Bright Line falls is owned by FD-BL (§4.2) for the industrial corpus and by FD-SBL for the surveillance corpus. Any instrument that states, cites, or restates the determinism or layer boundary as the *placement criterion* varies from this instrument and shall be conformed. This instrument supersedes the Layer Decomposition and Determinism Terminology Annex v0.1; references to that annex resolve here.

2.2 Family membership

FD-LD qualifies as a Foundational Definition: the correctness of the sector frameworks, the facility-class instruments, and the Command Broker Framework depends on agreeing what kind of thing the determinism boundary is — a scope gate, not a placement line.

2.3 Declaration convention

Per the family convention, the ontology is declared first (§4.1): the determinism boundary is a **scope boundary**. It partitions components by whether their behaviour has been shown to be bounded by proof, in order to decide which components are subject to Bright Line analysis. It is not itself the Bright Line and not the criterion for where the Bright Line falls.

2.4 What this instrument does not alter

The annex's equation of the layer boundary with the Bright Line placement criterion — its third mapping-table row in its original form — is withdrawn. The annex's canonical wording of the carve-out is no longer governing: the governing statement of scope is at §4.3, on the author's determination of 2 October 2026, and the earlier wording is kept as history in the Record. Everything else the annex held is carried. The Record sets out what the v0.1 promotion dropped and v0.2 restored.

3. The Two Vocabularies, One Scope Boundary

The table is retained from the annex. Its header row is this instrument's, and its third row is carried in repaired form (see the Record).

ISA / design-basis vocabulary	Determinism vocabulary	The single underlying property
Infrastructure layer	Deterministic systems; deterministic components; deterministic machine learning within a stated input bound and formally testable as deployed	Behaviour is reproducible and formally verifiable. The output space can be bounded and exhaustively reasoned about. Existing standards — ISA/IEC 62443, IEC 61508, IEEE 1012 — are adequate to it, except in the coverage gap recorded at §4.8.
Cognitive layer	Generative AI; non-deterministic systems; Gen AI; any component not shown to be deterministic within a stated input bound and formally testable as deployed	The same input may produce different outputs across execution cycles, or determinism over a stated bound has not been established and formally tested. The response space cannot be, or has not been, exhaustively enumerated. Formal verification does not apply or has not been carried out, and empirical assessment is required (see §4.8 on the assessment gap).
The boundary between the layers	**Not the placement criterion.** The scope boundary for Bright	Whether a component's behaviour can be bounded by

	Line analysis (§4.3; cited by FD-BL §4.4) — which components' actions are subject to the threshold	proof. This decides **scope** — whether a component is in or out of Bright Line analysis — and the assessment method that applies. It does **not** decide placement, which is where an in-scope component's individual actions fall and is drawn on consequence, action by action, per FD-BL §4.2–§4.3.

The infrastructure/cognitive distinction and the deterministic/non-deterministic distinction are the same boundary in two vocabularies. That boundary is a **scope boundary**: which side a component falls on decides whether it is subject to Bright Line analysis and the Command Broker obligation.

The protective functions FD-LS defines as locally sovereign sit on the infrastructure side of this boundary. FD-LS §3.2 states the relation — every locally sovereign function is a carved-out component, and not every carved-out component is locally sovereign — and this instrument adds no second layer vocabulary for it.

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4. The Definition

4.1 What kind of thing the determinism boundary is

The determinism boundary is a scope gate. It partitions components by whether their behaviour has been shown to be bounded by proof, and it decides which components are subject to Bright Line analysis. A component whose behaviour is deterministic within a stated input bound and formally testable as deployed is on the infrastructure side and outside the analysis. Every other component is on the cognitive side and within it, including a component whose determinism has not been established over a stated bound or has not been formally tested (§4.3).

The boundary is not the Bright Line, and which side a component falls on does not decide where the line falls for that component's actions. It decides only whether those actions are analysed under the line at all.

4.2 The correction: the layer boundary is not the placement criterion

The annex asserted that the layer boundary *is* the Bright Line placement criterion, and that Command Broker interposition follows from which side a component falls on. That assertion is withdrawn.

Determinism decides scope. Consequence decides placement (FD-BL §4.2). Fusing them produces two errors at once, both named in FD-BL §4.4: it implies that every action of a non-deterministic component is above the line — including actions of no material consequence — and it implies that a deterministic component can never produce an above-line action, which is false and which removes from analysis the automated systems most capable of causing harm.

An in-scope component that alters a protective function's settings, trip logic, bypass state or actuation criteria takes an action upon that function. The action is placed on the consequence of the alteration

(FD-BL §4.2), and the carve-out the protective function enjoys does not extend to it (FD-BL §4.4). A carved-out component that makes the same alteration is a different case. It is outside the obligation by the carve-out, and its action is not thereby below the line. That case falls in the coverage gap recorded at §4.8, and neither this instrument nor FD-BL resolves it.

Command Broker interposition therefore does not follow from "which side a component falls on." It follows from two separate determinations in sequence: whether the component is in scope (this instrument), and whether a given action is above the line on its consequence (FD-BL §4.2, action by action per §4.3).

4.3 The machine-learning carve-out

The governing statement. FD-LD owns the scope rule, and FD-BL §4.4 cites it from this section. The carve-out sets scope — which components are subject to Bright Line analysis — and it is not the line.

> Bright Line analysis and the Command Broker obligation apply to every component that does not meet the carve-out.
>
> A component meets the carve-out only if it is deterministic within a stated input bound and formally testable in the deployed configuration. Both conditions must hold, and the first is not severable from its bound.
>
> Generative and other non-deterministic components are the principal cases within scope. They are not its definition: a deterministic component never formally tested, or a generative model run at a fixed seed, is within scope because it does not meet the carve-out.
>
> The carve-out removes a component from this obligation only. It is not a grant of ungoverned autonomy, and it is not a finding that the component's actions are below the line.
>
> An action produced while a carved-out component's inputs lie outside the stated bound is not covered by the carve-out. How that condition is detected and the action routed is open at this version.
>
> An action produced while a carved-out component runs a configuration other than the one that was formally tested, including a configuration altered outside a governed act (*Local Sovereignty* §3.2), is not covered by the carve-out. How that condition is detected and the action routed is open at this version.

This distinction is not a nuance. A great deal of machine learning already deployed in critical infrastructure — anomaly detection, predictive maintenance, pattern-recognition algorithms, condition-based monitoring — is deterministic within bounded input ranges, produces bounded and testable outputs given bounded inputs, and is adequately governed by the existing standards ecosystem. Extending cognitive-layer assessment to it would impose unworkable requirements on tools that are already in service and functioning safely. **The carve-out exists to protect that population**, and an instrument that states the test without stating what the test is for invites a reader to apply it more widely than it was meant.

Where obligations on carved-out components arise. Verification and qualification obligations for deterministic components arise outside this obligation: in the functional safety ecosystem, in the industrial automation security ecosystem, and for nuclear digital instrumentation and control in the safety system design and qualification standards applicable to that sector. This instrument names the ecosystems and does not cite clauses within them.

A component is exempt from cognitive-layer assessment only if it is deterministic within a **stated input bound** and formally testable in the deployed configuration. The qualifier is essential **and it is not severable**: determinism is conditional on the input range, and outside that range the guarantee does not hold. An exemption asserted as "merely deterministic" without tying the claim to the range over which determinism was established is not a valid carve-out.

The test binds in both directions. A component is not exempted by being described as machine learning; a model whose outputs are not reproducible under identical inputs is a cognitive-layer system whatever its architecture is named. Any component relying on the carve-out shall have its input bound stated and verified as part of assessment.

The carve-out shall not be narrowed to enumerated roles. An instrument that permits deterministic machine learning to operate autonomously in named roles has converted a property test into a use-case list, which will exclude conforming components for reasons unrelated to their behaviour and will invite relabelling by components that do not conform.

Deterministic protective functions — reactor trip, engineered safety-feature actuation, protective-relay operation — are already automated and deterministic. They are outside the Command Broker obligation, but **by the carve-out (scope), not by being below the line.** Asserting they are below the line would be false, because they are among the highest-consequence actions in any facility; they are outside the obligation because their behaviour is bounded by proof (FD-BL §4.5). This distinction is the whole point of separating scope from placement, and it is why the fused row had to go.

Why the broker requirement is cross-sector. It derives from a property — behaviour not shown to be bounded by proof — of which the non-determinism of Generative AI systems is the principal case, and not from any feature of a particular industry. The property is sector-independent. A sector adopting the requirement is not importing another sector's practice; it is meeting a property of the technology that its own sector shares. This belongs in the definition because the alternative reading — that the requirement is a nuclear or an electric convention being generalised — is the one a sector reaches for when it wants to argue the requirement does not apply to it.

4.4 Scope, placement, and obligation are three questions

FD-LD owns the first. Which components are analysed at all is scope (this instrument). Where the line falls for an analysed component's actions is placement, on consequence (FD-BL §4.2). Who authorises an above-line action is the obligation, discharged by the Broker (FD-BR). No one of the three answers another.

Cross-corpus note. FD-LD supplies the determinism and layer vocabulary to both corpora. The carve-out — components that meet it are out of scope — applies to the industrial Bright Line. FD-SBL §4.4 explains why the carve-out does not transfer to the surveillance Bright Line: a deterministic statistical model's societal effect is not bounded by proof, so determinism is not a valid scope gate there. FD-LD's vocabulary is shared; its carve-out is industrial.

4.5 The term *two-layer architecture*, and what it does not denote

This instrument owns the layer vocabulary, so the reservation of the term belongs here and nowhere else.

The term *two-layer architecture* is reserved, across this series, for the infrastructure/cognitive decomposition. It does not denote the classical formal-methods pairing of an application layer and an

infrastructure layer, which is a decomposition of the deterministic stack alone and in which **both layers are deterministic**.

That classical pairing is not a competing model. It describes the internal structure of the infrastructure layer, and the table below sets that structure out with the cognitive layer beside it for contrast:

Layer	Contains
Infrastructure layer (deterministic)	Control logic and algorithms. **Platform:** operating systems, communication protocols, data storage, input/output. **AI-adjacent deterministic components:** retrieval against a known corpus, input validation, and deterministic machine learning within a stated input bound and formally testable as deployed.
Cognitive layer (non-deterministic, or not shown to be deterministic and formally testable)	Generative AI components. Confidence scoring emitted by the model. Any model whose outputs are not reproducible under identical inputs. Any component not shown to be deterministic within a stated input bound and formally testable as deployed.

Accordingly the lower level of the classical pairing is called the *platform level*, not the infrastructure layer — which now names the whole deterministic domain and would otherwise be nested inside itself.

The term *application layer* is not used for any verification-layer purpose. In industrial control usage it denotes the control application; in product architecture it commonly denotes a business-logic tier. Neither meaning aligns with the verification boundary, and the term should not be pressed into that service.

4.6 A product architecture tier is not a verification layer

A product architecture tier is not a verification layer, and the two should never be assumed to coincide. A single product tier may hold deterministic business logic and workflow orchestration alongside generative query routing — in which case it **straddles both verification layers**, and an assessor who maps the tier onto a layer will assess half of it and leave the other half unassessed.

Layer assignment is therefore made per component, with the basis recorded, and an architecture diagram is not a substitute for that assignment.

This is the operational consequence of §4.1. A scope gate that partitions components must be applied to components; applied to a drawing, it silently exempts whatever the drawing grouped with something deterministic.

4.7 Relationship to the dual design basis

This section is placed inside the definition rather than after it because it is a restatement of the definition rather than a consequence drawn from it.

A facility may carry two design bases at once: a **physical Design Basis** governing its plant and safety systems, and an **AI Governance Design Basis** governing the conditions under which an in-scope component may influence them.

In this instrument's vocabulary, an AI Governance Design Basis is a design basis for the cognitive layer. It is required precisely because that layer cannot be bounded by proof. The infrastructure layer beneath it — including deterministic machine learning within a stated input bound and formally testable as deployed, and including a plant's own automated protective actions — is governed by the physical Design Basis and the existing standards ecosystem, and **does not require an AI Governance Design Basis at all**, subject to the coverage gap recorded at §4.8, which this instrument names and does not close.

Stated that way, the dual design basis and the two-layer decomposition are the same claim seen from two directions.

4.8 Gaps recorded, not closed

This instrument records two gaps rather than papering them.

The coverage gap for deterministic machine learning. Deterministic machine learning taking autonomous, consequential control action — automated dispatch, voltage regulation, load allocation — sits in a partially covered area of the existing standards ecosystem. The coverage is stronger for conventional deterministic protective functions, which have decades of settled treatment, than for machine-learned components that satisfy the property test. A carved-out component that alters a protective function's settings (§4.2) lies in this gap. Closing it is outside the scope of this instrument and outside the current scope of this corpus, and it is identified here so that the carve-out is not read as an assertion that no gap exists.

The assessment-method gap for non-generative in-scope components. Stating scope as the complement of the carve-out (§4.3) brings into scope components that the corpus's assessment methods were not built for: a stochastic control policy, or a deterministic network that has never been formally tested. The cognitive error classes and sample-size logic of the Tiered Assessment Framework, and the Generative-AI-specific competencies of the nuclear Command Broker annex, are shaped by generative behaviour. Such a component carries the Command Broker obligation without an assessment method built for it. The gap is named here so that it is not hidden, and it is not closed at this version.

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5. Worked Examples

A supervised anomaly-detection model on a feedwater pump, deterministic within a stated input bound and formally testable as deployed, is an infrastructure-layer component: outside Bright Line analysis (§4.3). Replace it with a generative system producing the same recommendations, and it becomes a cognitive-layer component: within scope, its actions then placed on consequence (FD-BL §4.2).

The same anomaly-detection model receives inputs outside its stated bound after a sensor is replaced and its range changes. Actions the model produces while that condition holds are not covered by the carve-out (§4.3). How the condition is detected and those actions routed is open at this version (§7).

A vendor asserts a component is “deterministic” but cannot state the input range over which that holds. The assertion fails the qualifier (§4.3); the component is treated as cognitive-layer until the bound is stated and verified.

A reactor-trip function performs an action of the highest consequence without a Command Broker. It is not below the line; it is outside the obligation by the carve-out, because its behaviour is bounded by proof (§4.3). This is the example that makes the scope/placement distinction unavoidable.

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6. Conformance

The instruments that inherited the fused doctrine from the superseded annex conform to FD-LD by separating scope from placement:

DBA-DC-L1 §3.5 (series root), **DBA-EN-SF**, **DBA-ES** — restate the determinism boundary as the scope gate and place the line on consequence per FD-BL §4.2. Detailed diagnosis and replacement text are in FDBL-CS-P2.

Command Broker Framework §3.2 — resolve the layer/consequence contradiction in favour of consequence; the determinism boundary is scope. Markup drafted.

ISA Parent Framework — the annex was issued under its consistency protocol; update the reference to cite FD-LD and confirm the ISA Parent’s own placement language (Pass 2 of the coherence scan).

Instruments carrying the superseded carve-out wording — any instrument that states scope as “Generative AI only,” or says that carved-out machine learning “may operate autonomously,” conforms at its next revision to the governing statement at §4.3, citing this section rather than restating it.

The annex itself is superseded and withdrawn. One copy of it is still live and is not this instrument; it is described in the Record.

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7. Author Decisions Outstanding

Whether the “two vocabularies, one boundary” unification (§3) is retained as drafted — scoped to the scope boundary — or restructured. It is kept here because it is correct and useful once placement is removed from it.

The ISA Parent Framework consistency-protocol clause under which the annex was issued should be re-pointed at FD-LD, and the ISA Parent’s own statement of the boundary confirmed against this instrument.

Open at this version: how a carved-out component’s inputs lying outside the stated bound, or its running a configuration other than the one formally tested, is detected, and how actions produced under either condition are routed (§4.3). The principle is stated; the mechanism is not.

Closed in the October 2026 Edition: the designation FD-LD and its filing with the Foundational Definitions family. FD-LD has been a member of that family since the August 2026 Edition, and the October 2026 Edition is its third. The decisions closed at v0.3 are kept in the Record.

8. References

Foundational Definitions are cited here by designation, without version tokens. FD-REG holds the version state of FD-BL and FD-BR; FD-SBL-REG holds that of FD-SBL.

FD-BL — *The Bright Line*. Eclectic Technologies, 2026. (Owns placement; cites this instrument's §4.3 for the scope rule.)

FD-SBL — *The Surveillance Bright Line*. Eclectic Technologies, 2026. (§4.4 explains why the determinism carve-out does not transfer to the surveillance domain.)

FD-BR — *The Broker*. Eclectic Technologies, 2026.

Layer Decomposition and Determinism Terminology Annex. Version 0.1. Eclectic Technologies, 2026. (Superseded by this instrument.)

Roxey, T. E. *Threads to Formal Methods — a Grand Challenge*, with Appendix B. Eclectic Technologies, 2025. (Provenance only: the first statement of the division between provable software components and cognitive components. See R.7.)

Roxey, T. E. *Hybrid Formal Methods for Generative AI in Industrial Control Systems*. Version 6. Eclectic Technologies, 2025. (Provenance only: the first use of *infrastructure layer* and *two-layer architecture*, in the classical sense §4.5 departs from. See R.7.)

ISA/IEC 62443; IEC 61508; IEEE 1012 — the standards named at §3 as adequate to the infrastructure layer, subject to §4.8.

Record

This Record carries the provenance of this instrument. It is not part of the definition, and nothing in it varies §§1–8.

R.1 What the v0.1 promotion dropped, and what v0.2 restored

Until v0.4 the following stood at §2.4, after a paragraph stating that the placement equation was the only withdrawal from the annex and that at v0.2 everything else the annex held was carried for the first time. It is carried here unchanged; "this section" in it means the former §2.4.

At v0.1 it was not. This section then read that the two-layer decomposition, the determinism vocabulary, the machine-learning carve-out and the treatment of deterministic protective functions were *inherited from the annex unchanged in substance*, and that *only* the placement equation was withdrawn. **That was a completeness claim about a set the promotion never enumerated.** The promotion was a rewrite into Foundational Definition form, not an edit of the annex, and a rewrite carries what its author is looking at. Eleven further pieces of the annex did not carry over, in six classes, and none of them had anything to do with the defect the promotion was hunting:

Three whole sections. *The Two-Layer Architecture and What It Is Not*, *A Note on Product Architecture*, and *Relationship to the Dual Design Basis* — restored at §4.5, §4.6 and §4.7.

The canonical statement of the carve-out. The annex reproduced it as a block quotation, marked *as governing*, and named it the source text FD-BL §4.4 inherits. v0.1 paraphrased the carve-out's test and did not carry the statement, so the instrument that owns the carve-out did not contain its canonical wording. Restored at §4.3.

The carve-out's justification. The paragraph opening *this distinction is not a nuance* — the deployed machine learning it protects (anomaly detection, predictive maintenance, pattern recognition, condition-based monitoring) and the reason for protecting it. Restored at §4.3.

Two load-bearing clauses. That the *within bounded inputs* qualifier **is not severable**, which is a separate claim from its being essential; and the derivation of the broker requirement's cross-sector reach from non-determinism as a **sector-independent property** rather than from any feature of an industry. Restored at §4.3.

Three concordance terms. *deterministic components* and *Gen AI* as recognised synonyms, and **IEEE 1012** in place of a bare *IEEE*. A concordance exists so that a reader arriving in one vocabulary recognises the other, so dropping synonyms costs it exactly what it is for. Restored at §3.

The naming of the two document families. The annex said which documents speak which vocabulary — the ISA Parent Framework and the instruments derived from it on one side, the Command Broker family and the Design Basis Accident instruments on the other. v0.1 named the two vocabularies and not their populations, so a reader could learn that the boundary has two names without learning which name to expect in the document in front of them. Restored at §1.

This eleventh piece was itself missed once. The figure this section carried when it was drafted was *ten, in five classes*, and it was wrong because the enumeration was written from a four-term search of v0.1 rather than from the paragraph-by-paragraph comparison that produced the restorations. **It was caught by reading this instrument through before it left the project**, which is the practice registered as CR-TA at Corpus_Rules_Candidate_Register §9R. The corrected figure is eleven, and the enumeration above is now the comparison rather than a summary of it.

This is recorded rather than quietly repaired, because the claim stood in the corpus for a month inside a Foundational Definition, because instruments have been conformed against this instrument in that time, and because the shape is registered as CR-CC in Corpus_Rules_Candidate_Register §9S with this section as its second instance.

The restored material sits, at this version, in §1 (the naming of the two document families), §3 (the three concordance terms), §4.3 (the justification and the two load-bearing clauses; the canonical statement is now history at R.3), and §4.5, §4.6 and §4.7 (the three sections). Each carried a *restored at v0.2* tag in v0.2 and v0.3; the tags were moved here at v0.4.

One sentence of §1 carried the same defect. Its v0.1 text read that the instrument *withdraws the fused row, keeps everything else*, and that was not true of v0.1.

R.2 The concordance's third row

Until v0.4, §3 opened by saying that its header row is this instrument's, that at v0.2 the cell terms were restored to the annex's, and that at v0.3 the third row returned in repaired form. It added that the v0.1

statement that the table was *corrected only in its header row and in the removal of the placement equation* was itself inexact, because three terms had been dropped from the cells in the same pass. The following note stood beneath the table; "above" in it means the §3 table.

On the third row, which has been in three states. The annex's original row added *The Bright Line placement criterion* to this equivalence, and that form is withdrawn and stays withdrawn (§4.2). v0.1 withdrew the row entirely. **v0.3 restores it in the repaired form above, on the author's determination of 13 September 2026**, because a table is a place readers arrive at directly and the two-row form stated the equivalence while saying nothing about what the boundary decides — leaving the reader to find that three sections away at §4.1 and §4.2, which is the arrival problem CR-VA exists to name.

The middle cell leads with the negation deliberately. A row in this table is the most copied object in the instrument; the fused form propagated as doctrine into four instruments precisely because it was copyable. **A reader who copies only the opening clause of the repaired row copies *not the placement criterion*, which is the half that matters.**

R.3 The Command Broker family's wording of the carve-out, kept as history

From v0.2 to v0.3 this instrument reproduced the following, as governing, in §4.3. The wording is the Command Broker family's, and FD-BL §4.4 was then described as inheriting it.

> Machine learning systems that are deterministic within bounded inputs and formally testable may operate autonomously without Command Broker interposition. The Bright Line and the Command Broker requirement apply to Generative AI only.

It is no longer governing. On 2 October 2026 the author determined that scope is stated as the complement of the carve-out (Decision B, Option B3, in the Action Plan for the October 2026 Edition). The governing statement at §4.3 replaces this wording, removes "may operate autonomously," and attaches the guard that the carve-out is not a grant of ungoverned autonomy. The two reasons are that "Generative AI only" names a technology class while the corpus grounds the obligation in a property, and that the narrow form leaves out a component that is deterministic but never formally tested, which is neither generative nor carved out.

R.4 The disclosed edit to the §4.5 lead-in

Until v0.4 this note stood at the end of §4.5; "above" in it means the §4.5 lead-in.

One edit is disclosed. The annex introduced the table with *it describes the internal structure of the infrastructure layer*, while the table carries a cognitive row as well. The lead-in above says what the table does. The table itself is carried unchanged.

R.5 The back-ported copy of the annex

Until v0.4 the following stood in §8.

A back-ported copy of the annex, Layer_Determinism_Terminology_Annex_v0_1.docx, sits at Project 2's working root. Its file name carries the annex's title and v0.1; the document inside heads itself *FD-LD … Version 0.2 — DRAFT* and carries a repaired third concordance row, while its closing line still reads *Version 0.1 — DRAFT … this annex reconciles vocabulary; it establishes no governance concept and modifies no requirement*. **It is neither a successor to this instrument nor a stale copy of the annex**: it is the v0.1 correction applied to the annex's fuller body, made outside this project and never returned, and it is the text the restorations at §2.4 were recovered from. **Its FD-LD designation has no standing** —

the family and its register are this project's, and FD-REG governs the version state. Recorded here because a reader who meets that file needs to know what it is.

Its disposition was determined on 13 September 2026. A copy is in this project's _archive/ as Layer_Determinism_Terminology_Annex_BACKPORT_recovered_2026-09-13.docx, hash-verified against the original, and the original is on the author's manual list for removal from Project 2's working root. **The archived copy is the provenance record for the eleven restorations at §2.4** — the evidence that the losses were real and of what the recovered text said.

R.6 Decisions closed at v0.3

Until v0.4 these stood in §7. "See §3" in them means the §3 table and R.2; "See §8" now means R.5.

Closed at v0.3: the concordance's third row returns in repaired form, with the negation leading. Determined by the author on 13 September 2026 and recorded at Determination_2026-09-13_Six_Author_Decisions_On_The_Pending_Sheet §2. The reasoning both ways is in _sessions/Author_Decisions_Pending_2026-09-13_Project_0 §2 and is not restated here. See §3.

Closed at v0.3: the back-ported copy goes to Project 0's _archive/, under a name that says what it is. Determined the same day and recorded in the same place. It is Foundational Definitions material, this project owns the family, and the file is the provenance record for the eleven restorations v0.2 made — which is the record that belongs with the family rather than in a project that cannot issue one. See §8.

R.7 Before the annex: the formal-methods papers of 2025

The division this instrument defines was first stated in the author's *Threads to Formal Methods — a Grand Challenge* series, written in 2025 to propose a national effort on quality assurance for generative AI in industrial control systems. Its Appendix B, in the version whose file is dated 3 September 2025, states that generative AI is fundamentally a system of systems: some systems handle inputs and outputs for a cognitive component supported by machine learning, where inference occurs. It concludes that formal methods apply to the software parts and a different assurance approach to the cognitive components. The passage stays in every later version of the appendix and enters the main paper in the version dated 19 October 2025, where it stands unchanged through the version dated 28 October 2025 (v16). This is the scope boundary of §4.1 in its first form: a partition of a system by whether its behavior can be bounded by proof, with a different method of assurance on each side.

The vocabulary came later. The terms *infrastructure layer* and *two-layer architecture* first appear in *Hybrid Formal Methods for Generative AI in Industrial Control Systems* V6, whose PDF is dated 28 November 2025. There the two-layer architecture meant the classical formal-methods pairing of an application layer, holding the control logic, over an infrastructure layer of operating system and network stack, both deterministic, with the cognitive components operating on top of that foundation as a separate concern. *Layer Decomposition and Determinism* §4.5 reserved the term for the infrastructure/cognitive decomposition instead, renamed the lower level of the classical pairing the platform level, and stated that the reserved term does not denote the classical pairing. The reservation is therefore a deliberate departure from the term's first use in this body of work, and §4.5 is the record of it.

The papers are cited at §8 as provenance. They are not authority for anything in §§1–7.

Revision History

0.5 — 15 October 2026. Issued in the November 2026 Edition, and recorded in FD-REG v0.32 in the same act. Two edits queued in FD-REG. First (FD-REG v0.25), from Determination_2026-10-08_Untested_Logic_Is_Not_Covered_by_the_Carve-Out, option (b): §4.3 gains a sentence beside the out-of-bound sentence, that an action produced while a carved-out component runs a configuration other than the one formally tested, including one altered outside a governed act, is not covered by the carve-out, citing *Local Sovereignty* §3.2 for *governed act* at its first use here rather than restating it; and the open item at §7 widens to take in that condition. Second (FD-REG v0.29), from FD-LD_Queued_Edit_for_the_November_Cycle_Where_the_Layer_Split_Began v0.2, approved 14 October 2026: the Record gains R.7, tracing the layer split to the author's 2025 formal-methods papers, and §8 gains two provenance-only references. No definition changes beyond the first edit; no section is renumbered.

0.4 — 2 October 2026. Issued in the October 2026 Edition, and recorded in FD-REG v0.22 in the same act. One cumulative entry, extended at each step of the Action Plan for the October 2026 Edition that opens this instrument (Steps 2 and 6).

Step 2, 2 October 2026 (F-02, F-06; F-15, F-16, F-20, F-24). §4.3 carries a new governing statement of scope on the author's determination of 2 October 2026 (Decision B, Option B3): Bright Line analysis and the Command Broker obligation apply to every component that does not meet the carve-out, generative and other non-deterministic components being the principal cases and not the definition. "May operate autonomously" is removed, the guard that the carve-out is not a grant of ungoverned autonomy is attached, and an action produced while inputs lie outside the stated bound is stated not to be covered, with the mechanism left open (§7). The Command Broker family's wording is kept as history at R.3. §4.3 also takes from FD-BL §4.4, which now cites this section, the prohibition on narrowing the carve-out to enumerated roles and the naming of the ecosystems where obligations on carved-out components arise. The §4.3 cross-sector paragraph grounds the requirement in the property rather than in Generative AI as a class. §2.3 and §4.1 state the partition as *shown to be* bounded by proof, and the cognitive-layer rows of the §3 and §4.5 tables include any component not shown to be deterministic and formally testable. **F-06:** §4.2 now states the actions-upon clause with an in-scope actor, matching FD-BL §4.4, and says that a carved-out actor altering a protective function's settings falls in the recorded gap, which neither instrument resolves. **F-20:** new §4.8 carries the coverage gap for deterministic machine learning, moved from FD-BL §4.4, and the assessment-method gap that Decision B opens; the §3 infrastructure cell and §4.7 are qualified by it. **F-15:** §8 drops the version tokens on FD-BL, FD-SBL and FD-BR and adds a lead-in naming FD-REG and FD-SBL-REG. **F-16:** §7's first item, the designation, is closed. **F-24:** provenance narrative moves out of §§1–8 into a new Record section before this history: the §2.4 account of the v0.1 losses (R.1), the note on the third row (R.2), the superseded carve-out wording (R.3), the disclosed §4.5 edit (R.4), the back-ported copy (R.5) and the decisions closed at v0.3 (R.6). The *restored at v0.2* tags are removed and R.1 says where each restored piece now sits. §2.4 is rewritten to state the two changes to annex content. §1's account of the fused doctrine is put in the past frame. A worked example for out-of-bound inputs is added at §5, and §6 adds a conformance line for instruments carrying the superseded wording. No section is renumbered except by the addition of §4.8 and the Record. FD-LS's §3 and the F-21 concordance wait for Step 6.

Step 6, 2 October 2026 (F-21 on FD-LD's side). §3 gains a paragraph after the concordance placing FD-LS's locally sovereign functions on the infrastructure side of the boundary and pointing to FD-LS §3.2 for the relation, so that the family has one layer vocabulary. FD-LS §4 is reworded in this instrument's terms rather than a second layer pair being added here.

Verification read, 2 October 2026. §7 counts the October edition as FD-LD's third. §4.4's cross-corpus note states the carve-out as components that meet it, not deterministic components. The §3 infrastructure cell and §4.7 say "within a stated input bound and formally testable as deployed," and §4.7 speaks of an in-scope component rather than a non-deterministic system. The §3 third row, the author's wording of 13 September 2026, is left for the author. No change to placement, granularity, scope or obligation.

Review of 2 October 2026. On the author's instruction to apply the cold review of the October 2026 Edition review draft as recommended (FD-BL Reconciliation/Independent_Review_2026-10-02_FD_October_2026_Edition_Review_Draft_v0_1), this instrument takes findings G-11, G-14; each edit is quoted in the review's application record.

0.3 — 13 September 2026. Restores the concordance's third row to §3 in repaired form, on the author's determination of this date recorded at Determination_2026-09-13_Six_Author_Decisions_On_The_Pending_Sheet §2. The row states that the boundary between the layers is **not the placement criterion** and is the scope boundary for Bright Line analysis under FD-BL §4.4, with the property cell distinguishing scope from placement explicitly. **The middle cell leads with the negation** rather than closing on it, on the author's direction, because a row in this table is the most copied object in the instrument and the fused form propagated into four instruments precisely by being copyable — so a reader who copies only the opening clause copies the half that matters. The annex's original fused form remains withdrawn (§4.2) and the row has now been in three states, which §3 records. Closes both author decisions opened at v0.2 (§7) and records at §8 that the back-ported copy is archived here, hash-verified, as the provenance record for the restorations. **No section was renumbered and no other text changed.**

One correction to the v0.2 entry below, made by annotation and not by edit. That entry opens *restores ten pieces of the annex*; the figure is **eleven**, as §2.4 of v0.2 itself states and enumerates. The count in §2.4 was corrected from ten to eleven during v0.2's terminal read, and the revision entry — written before that read and not re-read after it — kept the superseded figure. **The entry is not edited**, per CR-VA §4.1: an entry's assertions about its own version are annotated in a successor and never removed. **The defect is recorded as an instance under CR-TA**, whose §3 now states the limit it exposes: a repair made in response to the read reopens the read on what the repair touched.

0.2 — 13 September 2026. Restores ten pieces of the annex that the v0.1 promotion dropped, and corrects the three sentences that said nothing had been dropped. Adds §4.5 (*the term two-layer architecture, and what it does not denote* — the reservation of the term against the classical formal-methods pairing, the layer-contents table, the naming of the **platform level**, and the exclusion of *application layer* from verification use), §4.6 (*a product architecture tier is not a verification layer* — a tier may straddle both layers, so assignment is per component with the basis recorded), and §4.7 (*relationship to the dual design basis* — an AI Governance Design Basis is a design basis for the cognitive layer). Restores into §4.3 the **canonical statement of the carve-out** as a block quotation reproduced as governing, which v0.1 paraphrased and did not carry, so that the instrument owning the carve-out again contains the wording FD-BL §4.4 inherits; the *this distinction is not a nuance* justification with the deployed machine learning the carve-out protects; the **not severable** clause on the *within bounded inputs* qualifier; and the derivation of the broker requirement's cross-sector reach from non-determinism as a **sector-independent property**. Restores into §3 the three concordance terms dropped from the cells — *deterministic components*, *Gen AI*, and **IEEE 1012** for a bare *IEEE*. **Corrects the completeness claims at §1 and §3 and rewrites §2.4** to enumerate what was lost; the v0.1 entry below is annotated by this entry and is not edited, per CR-VA §4.1 as determined 13 September 2026. **No**

section was renumbered, existing citations to §2.1 and §4.x being load-bearing across the corpus. Opens two author decisions at §7 — whether the concordance's third row returns in repaired form, and the disposition of the back-ported copy at Project 2's working root — and records that copy at §8. **The losses are the second registered instance of CR-CC** (Corpus_Rules_Candidate_Register §9S): a rewrite carries what its author is looking at, and a sentence saying otherwise is cheaper to write than the comparison that would falsify it.

0.1 — August 2026. Promoted from the *Layer Decomposition and Determinism Terminology Annex* v0.1 to a Foundational Definition. Retained the two-layer decomposition (§3), the determinism vocabulary, the machine-learning carve-out (§4.3), and the deterministic-protective-function treatment. Withdrew the annex's third mapping row equating the layer boundary with the Bright Line placement criterion, and declared the determinism boundary a scope gate (§4.1) distinct from placement (§4.2), consistent with FD-BL §4.2/§4.4. Added the cross-corpus note (§4.4) that the carve-out is industrial and does not transfer to FD-SBL. This instrument removes, at its source, the conflation identified in the corpus coherence scan (FDBL-CS-SCAN-1) as the doctrinal origin of the placement error.